SOFTWARE REQUIREMENTS SPECIFICATION

IRO – Issues Reporting Outlet

A web-based system for reporting and resolving issues between
Adekunle Ajasin University, Akungba-Akoko (AAUA) students and the indigenes of
Akungba, Iwaro, Etioro, Ayegunle and neighbouring communities

Version 1.0

CSC426 Weekly Dev – Weekend 3 (01/07/26)
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Revision History

Version	Date	Description
1.0	July 2026	Initial SRS for the IRO mini web interface.

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1. Introduction

1.1 Purpose

This document specifies the requirements of **IRO (Issues Reporting Outlet)**, a web-based system through which AAUA students and indigenes of the host communities report issues that affect them, and through which an IRO desk officer reviews, escalates and resolves those reports. It is written for the course lecturer, the project developer, and any future team that extends the system. It describes what the system must do (functional requirements), how well it must do it (non-functional requirements), and how its major components interact.

1.2 Scope

Relations between AAUA students and the host communities (Akungba, Iwaro, Etioro, Ayegunle and Oke-Oka) are affected by disputes over rent and accommodation, security, land, cultural festivals, sanitation, transport and prices. Today these issues are reported through scattered informal channels (the SUG, community chiefs, WhatsApp groups, the police), many are never reported for fear of victimisation, and reporters rarely learn what happened to their complaint.

IRO provides a single, neutral outlet that:

- lets any student or indigene submit an issue report, anonymously if they wish;
- gives every report a unique ticket ID that the reporter uses to track its progress;
- gives the IRO desk officer a dashboard to view, filter, update, escalate and close reports;
- keeps a timestamped history of every status change for accountability.

Out of scope for version 1.0 (the mini interface): SMS/e-mail notifications, file/photo uploads, a server-side database, multiple officer accounts and Yoruba translation. These are listed as future enhancements in Appendix B.

1.3 Definitions, Acronyms and Abbreviations

Term	Meaning
IRO	Issues Reporting Outlet – the system specified here.
AAUA	Adekunle Ajasin University, Akungba-Akoko, Ondo State, Nigeria.
Indigene	A native or long-term resident of Akungba, Iwaro, Etioro, Ayegunle, Oke-Oka or neighbouring communities.
Reporter	Any user (student, indigene or other) who submits a report.
Desk Officer	The administrator who manages reports on the IRO desk.
Escalation Authority	A body to which a report is forwarded: Students' Affairs, SUG, community leaders (Oba/Baale), campus security or police.
Ticket ID	Unique report reference in the form `IRO-YY-XXXXX`, e.g. IRO-26-4F7K2.
SRS	Software Requirements Specification.
FR / NFR	Functional requirement / non-functional requirement.
UI	User interface.

1.4 References

- IEEE Std 830-1998, *IEEE Recommended Practice for Software Requirements Specifications*.
- IRO Community Needs Questionnaire (Google Form) – `docs/questionnaire/`.
- IRO Ishikawa Problem Analysis – `docs/ishikawa/IRO_Ishikawa.drawio`.

- IRO Use-Case Diagram – `docs/use-case/IRO_UseCase.drawio`.
- Nigeria Data Protection Act (NDPA) 2023.

1.5 Overview

Section 2 describes the product in general terms. Section 3 shows the architecture and how the major components interact. Sections 4 and 5 list the functional and non-functional requirements. Section 6 covers external interfaces, Section 7 the use cases, Section 8 the data, and Section 9 traces the requirements back to the root causes found in the Ishikawa analysis.

2. Overall Description

2.1 Product Perspective

IRO is a new, self-contained web application. Version 1.0 is a static front-end (HTML, CSS, JavaScript) that runs in any modern browser and stores data in the browser's `localStorage`. All data access goes through a single data-layer object (`IssueStore`), so a later version can replace browser storage with a server API and database without changing the user interface.

2.2 Product Functions (summary)

- Present the purpose of IRO, reporting guidelines and issue categories.
- Capture an issue report with reporter type, category, severity, location, date, title and description.
- Allow anonymous reporting.
- Validate input and generate a unique ticket ID.
- Let reporters track status and history with the ticket ID.
- Let the desk officer log in, view statistics, search/filter reports, view details, change status with a note, escalate and delete reports.

2.3 User Classes and Characteristics

User class	Description	Technical skill
Student	Undergraduate or postgraduate student of AAUA, on or off campus. May supply a matric number.	Medium–high; mostly smartphone users.
Indigene	Landlord, trader, driver, farmer, chief or other resident of the host communities. Selects their community.	Low–medium; a simple, clear interface is essential.
Other reporter	Staff, visitors, traders not from the communities.	Varies.
IRO Desk Officer	Staff member (e.g. from Students' Affairs or a joint town–gown committee) who manages reports.	Medium; trained to use the dashboard.
Escalation Authority	Secondary actor that receives escalated issues offline; does not log in to v1.0.	N/A

2.4 Operating Environment

- Client: any modern browser (Chrome, Firefox, Edge, Safari, Opera Mini excluded) on Android, iOS, Windows, macOS or Linux.
- Server: none required for v1.0; the files can be opened locally or hosted on any static host (e.g. GitHub Pages).

2.5 Design and Implementation Constraints

- Must be simple to build and submit within one week (course deadline).
- Plain HTML5, CSS3 and JavaScript (ES6); no frameworks or build tools.

- Data in v1.0 is held in the browser, so reports are visible only on the device where they were made.
- Must respect the confidentiality of reporters in line with the NDPA 2023.

2.6 Assumptions and Dependencies

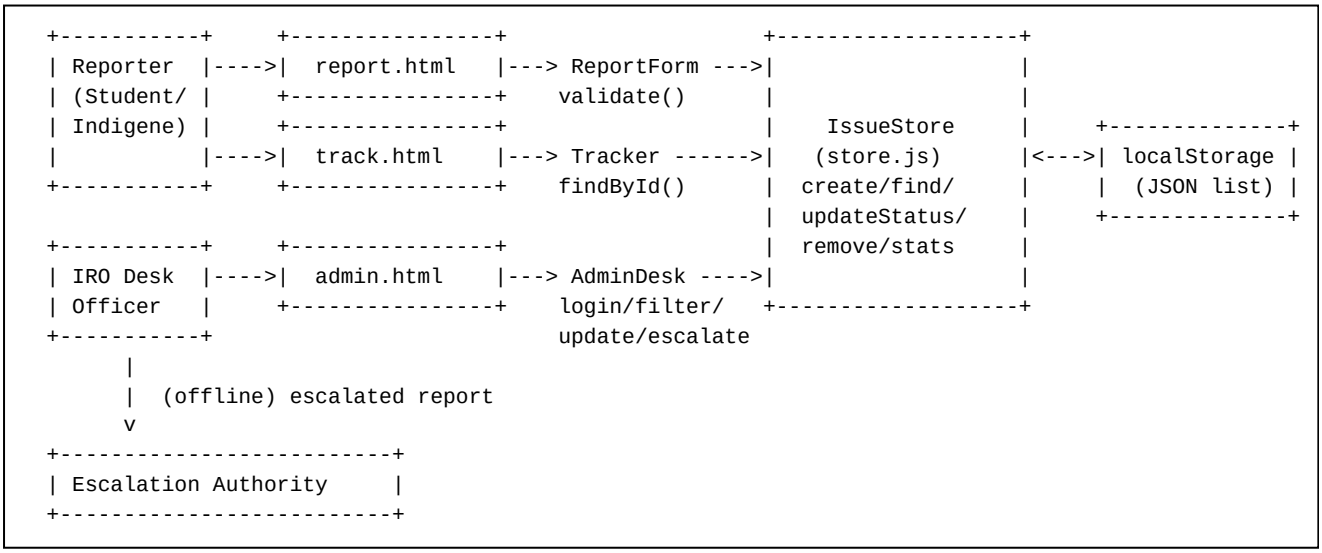
- Users have access to a browser with JavaScript enabled.
- The university and community leaders agree to act on escalated reports.
- A production deployment will add a server-side database and proper authentication.

3. System Architecture & Interaction of Major Components

3.1 Major Components

Component	Files	Responsibility
Presentation layer (pages)	index.html, report.html, track.html, admin.html, css/style.css	Displays the UI: home & guidelines, report form, tracking page and admin desk.
ReportForm controller	js/report.js	Shows/hides student or indigene fields, collects and validates input, calls IssueStore.create() and shows the ticket ID.
Tracker controller	js/track.js	Looks up a ticket ID via IssueStore.findById() and renders status and history (no personal data shown).
AdminDesk controller	js/admin.js	Handles login/logout, statistics, search/filter, detail view, status updates, escalation and deletion.
IssueStore (data layer)	js/store.js	Single point of access to data: create, read, find, update status, delete, statistics, ticket ID generation.
IRO configuration	js/store.js	Holds categories, communities, statuses, severities and the demo passcode.
Storage	Browser localStorage (key `iro_issues`)	Persists the list of issues as JSON. Replaced by a server database in future.

3.2 Component Interaction Diagram



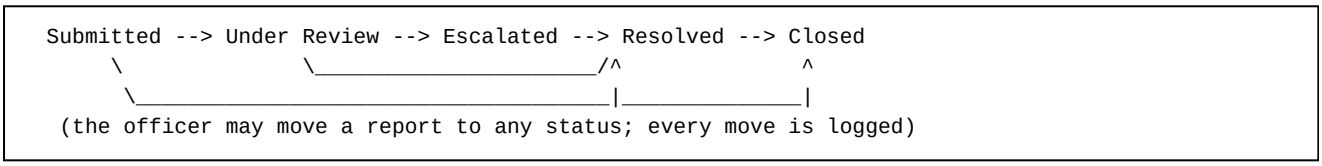
3.3 Key Interaction Sequences

(a) Submitting a report. Reporter opens report.html → selects reporter type (ReportForm shows matric field for students or community field for indigenes) → fills the form → ReportForm.validate() checks every field → IssueStore.create() builds the issue object, generates a ticket ID, sets status *Submitted*, adds the first history entry and saves to storage → ReportForm displays the ticket ID and a "Track this report" link.

(b) Tracking a report. Reporter enters the ticket ID on track.html (or follows the link) → Tracker calls IssueStore.findById() → if found, the title, category, severity, current status and full history are shown; otherwise an error message is shown. Name, contact and description are never shown on this page.

(c) Managing a report. Desk officer logs in on admin.html → AdminDesk loads IssueStore.stats() and IssueStore.getAll() → officer searches/filters and opens a report → officer chooses a new status (*Under Review*, *Escalated*, *Resolved*, *Closed*) and writes a note → IssueStore.updateStatus() records the change with a timestamp → the reporter sees the new status and note the next time they track the ticket. Choosing *Escalated* means the officer forwards the case to the relevant authority.

3.4 Report Status Life-Cycle



4. Functional Requirements

Priority: **H** = must have (in v1.0), **M** = should have, **L** = could have (future).

ID	Requirement	Description	Pri.
4.1 Public information			
FR-01	Home page	The system shall display the purpose of IRO, the four-step reporting process, the list of issue categories and an emergency notice.	H
FR-02	Navigation	Every page shall have a navigation bar linking Home, Report an Issue, Track Report and Admin.	H
4.2 Issue reporting			
FR-03	Reporter type	The system shall require the reporter to choose Student, Indigene or Other.	H
FR-04	Conditional fields	If Student is chosen the system shall show an optional matric-number field; if Indigene is chosen it shall show a required community list (Akungba, Iwaro, Etioro, Ayegunle, Oke-Oka, Other).	H
FR-05	Anonymous reporting	The system shall let the reporter tick "Report anonymously"; the name and contact fields shall then be hidden and shall not be stored.	H
FR-06	Issue details	The system shall capture category, severity (Low, Medium, High, Emergency), location, incident date, short title and description.	H
FR-07	Validation	The system shall reject a report and show a clear message if a required field is empty, the date is in the future, the title is shorter than 5 characters, the description is shorter than 20 characters, or a contact given is not a valid phone number or e-mail.	H

FR-08	Ticket generation	On a valid submission the system shall create a unique ticket ID of the form IRO-YY-XXXXX (no ambiguous characters 0, O, 1, I) and set status to Submitted.	H
FR-09	Confirmation	The system shall display the ticket ID with a link to track the report.	H
4.3 Tracking			
FR-10	Track by ticket	The system shall let anyone enter a ticket ID (case-insensitive) and view the report's title, category, severity, date, current status and status history with notes.	H
FR-11	Privacy on tracking	The tracking page shall not display the reporter's name, contact, matric number or description.	H
FR-12	Unknown ticket	If no report matches, the system shall show "No report found with that ticket ID".	H
4.4 Desk officer (administration)			
FR-13	Login	The admin desk shall be accessible only after a correct passcode is entered; the session ends on logout or when the browser tab is closed.	H
FR-14	Dashboard statistics	The system shall show total reports, open reports, active emergencies and resolved reports.	H
FR-15	List, search & filter	The system shall list all reports newest first and allow search by ticket ID, title or location, and filtering by status, category and reporter type.	H
FR-16	View details	The system shall show a report's full details, including reporter identity (or "Anonymous") and history.	H
FR-17	Update status	The officer shall be able to set a new status with an optional note; the system shall append a timestamped history entry.	H
FR-18	Escalate	The officer shall be able to mark a report as Escalated with a note naming the authority it was forwarded to.	H
FR-19	Delete	The officer shall be able to delete a report after confirming (e.g. spam or duplicates).	M
4.5 Future functions			
FR-20	Notifications	The system should send SMS/e-mail alerts to reporters who give contact details when status changes.	L
FR-21	Evidence upload	The system should allow photo/video attachments.	L
FR-22	Reports & trends	The system should produce monthly summaries by category and community.	L

5. Non-Functional Requirements

ID	Attribute	Requirement
5.1 Performance		
NFR-01	Response time	Each page shall load and become usable within 2 seconds on a 3G mobile connection; form submission and search shall respond within 1 second for up to 1,000 stored reports.
NFR-02	Page weight	Total size of the app shall stay under 200 KB, with no external libraries, so it works on low-data plans.

5.2 Security & Privacy		
NFR-03	Confidentiality	Anonymous reports shall store no name or contact. Identifying data shall be visible only on the admin desk.
NFR-04	Access control	Admin functions shall require authentication. (v1.0 uses a demo passcode; production shall use server-side authentication with hashed passwords and HTTPS.)
NFR-05	Input safety	All user text shall be HTML-escaped before display to prevent script injection (XSS).
NFR-06	Ticket secrecy	Ticket IDs shall be random ($32^5 \approx 33$ million combinations per year) so they cannot easily be guessed.
NFR-07	Data protection	Handling of personal data shall comply with the Nigeria Data Protection Act 2023.
5.3 Usability & Accessibility		
NFR-08	Ease of use	A first-time user shall be able to submit a report in under 3 minutes without help.
NFR-09	Responsive design	The UI shall work on screens from 320 px (small phones) to desktop without horizontal scrolling.
NFR-10	Accessibility	All inputs shall have visible labels, keyboard focus shall be visible, and text contrast shall meet WCAG 2.1 AA.
NFR-11	Clear language	Messages shall be in plain English; Yoruba support is a future enhancement.
5.4 Reliability & Availability		
NFR-12	Data integrity	Every status change shall be logged; history entries shall never be edited or removed except when the whole report is deleted.
NFR-13	Fault tolerance	If stored data is unreadable the system shall start with an empty list rather than crash.
NFR-14	Availability	When hosted, the system shall be available 99% of the time (excluding planned maintenance).
5.5 Maintainability & Portability		
NFR-15	Modularity	All data access shall go through IssueStore so storage can be replaced without changing UI code.
NFR-16	Configurability	Categories, communities, statuses and severities shall be defined in one configuration object (IRO).
NFR-17	Portability	The app shall run in current versions of Chrome, Firefox, Edge and Safari with no installation.
NFR-18	Code quality	Source shall be commented and kept in a public GitHub repository with documentation.
5.6 Social / Ethical		
NFR-19	Neutrality	The system shall treat students and indigenes equally; the wording of forms shall not blame either group.

6. External Interface Requirements

6.1 User Interfaces

Screen	Main elements

Home (index.html)	Banner, "Report an Issue" and "Track My Report" buttons, 4-step process cards, categories, emergency notice.
Report (report.html)	Reporter type, matric/community, anonymous checkbox, name, contact, category, severity, location, date, title, description, Submit; success panel with ticket ID.
Track (track.html)	Ticket ID box, Track button, result card with status badge and timeline.
Admin (admin.html)	Login form; dashboard with 4 statistic cards, search and filters, report table, detail panel with history, status update form and Delete button.

6.2 Hardware Interfaces

None beyond a device capable of running a modern web browser.

6.3 Software Interfaces

Browser Web Storage API (localStorage, sessionStorage). Future: REST API to a server database (e.g. Node.js/PHP with MySQL), SMS gateway and e-mail service.

6.4 Communication Interfaces

HTTP/HTTPS when hosted. Production deployment shall use HTTPS only.

7. Use-Case Model

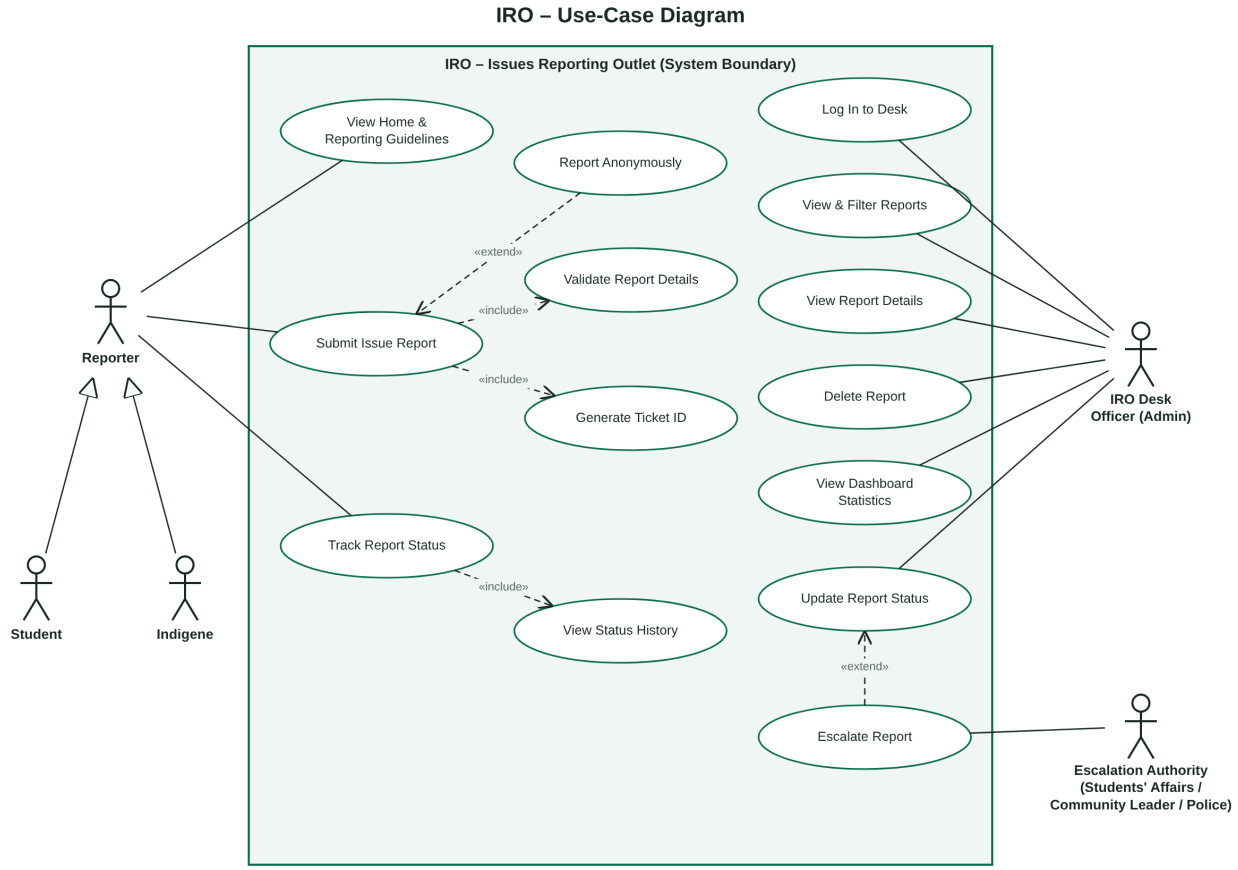


Figure 1: IRO use-case diagram (source: docs/use-case/IRO_UseCase.drawio)

7.1 Actors

- **Reporter** (generalised from **Student** and **Indigene**) – submits and tracks reports.
- **IRO Desk Officer** – manages reports.
- **Escalation Authority** – receives escalated reports (Students' Affairs, community leaders, police).

7.2 Use-Case Descriptions

Use case	Actor	Main flow	Alternative / notes
UC-01 View Home & Guidelines	Reporter	Reporter opens IRO and reads the process, categories and emergency notice.	–
UC-02 Submit Issue Report	Reporter	1. Choose reporter type. 2. Fill details. 3. Submit. 4. System validates (UC-03), generates ticket (UC-04) and shows it.	Validation fails → message shown, nothing saved.
UC-03 Validate Report Details	System («include»)	Checks required fields, date, lengths and contact format.	–
UC-04 Generate Ticket ID	System («include»)	Creates unique IRO-YY-XXXXX ID and first history entry.	–
UC-05 Report Anonymously	Reporter («extend» UC-02)	Reporter ticks Anonymous; identity fields are hidden and not stored.	–

UC-06 Track Report Status	Reporter	Enter ticket ID → system shows status and history (UC-07).	Unknown ID → error.
UC-07 View Status History	System («include»)	Displays every status change with time and note.	–
UC-08 Log In to Desk	Desk Officer	Enter passcode → dashboard opens.	Wrong passcode → error.
UC-09 View & Filter Reports	Desk Officer	Search by text, filter by status/category/reporter type.	No match → "No reports match".
UC-10 View Report Details	Desk Officer	Open a report to see all details and history.	–
UC-11 Update Report Status	Desk Officer	Choose new status, add note, save → history updated.	–
UC-12 Escalate Report	Desk Officer → Escalation Authority («extend» UC-11)	Set status Escalated with note naming the authority; case is forwarded.	Forwarding is offline in v1.0.
UC-13 Delete Report	Desk Officer	Confirm deletion → report removed.	Cancel → nothing changes.
UC-14 View Dashboard Statistics	Desk Officer	See totals of all, open, emergency and resolved reports.	–

8. Data Requirements

8.1 Issue Object (data dictionary)

Field	Type	Description / rule
id	String	Ticket ID, IRO-YY-XXXXX, unique.
reporterType	String	Student | Indigene | Other (required).
anonymous	Boolean	True if reporter chose anonymity.
name	String	Reporter's name; empty if anonymous.
contact	String	Phone or e-mail; empty if anonymous.
matricNo	String	Optional; students only.
community	String	Required for indigenes.
category	String	One of the configured categories.
severity	String	Low | Medium | High | Emergency.
location	String	Where the incident happened.
incidentDate	Date (YYYY-MM-DD)	Not in the future.

title	String (5–100)	Short summary.
description	String (20–2000)	Full account.
status	String	Submitted | Under Review | Escalated | Resolved | Closed.
createdAt	ISO date-time	Time of submission.
history	Array of {status, note, at}	Append-only log of status changes.

8.2 Configuration Lists

Categories: Security / Safety; Accommodation & Rent; Land & Property; Harassment / Assault; Cultural & Festival Conflicts; Sanitation & Environment; Water, Power & Utilities; Transport & Okada; Market & Pricing; Other.
Communities: Akungba, Iwaro, Etioro, Ayegunle, Oke-Oka, Other.

9. Problem Analysis (Ishikawa) and Requirements Traceability

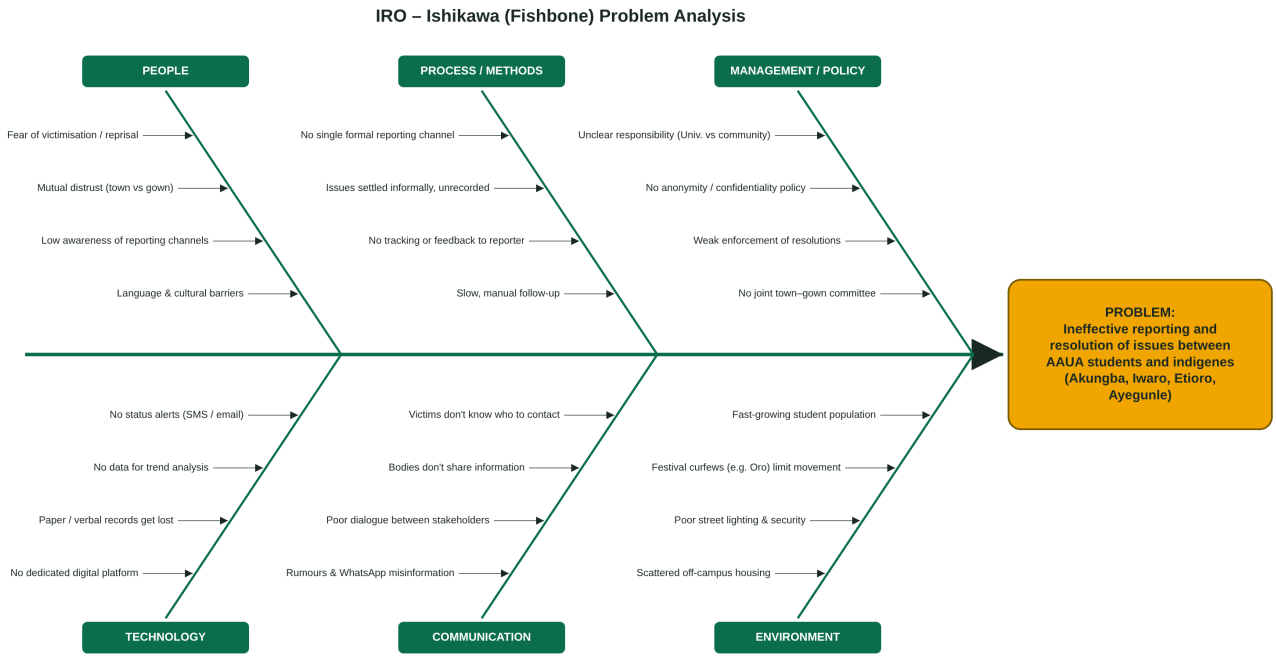


Figure 2: Ishikawa (fishbone) analysis of the problem (source: docs/ishikawa/IRO_Ishikawa.drawio)

Cause category	Root cause	Addressed by
People	Fear of victimisation / reprisal	FR-05, FR-11, NFR-03
People	Low awareness of reporting channels	FR-01, FR-02
Process / Methods	No single formal reporting channel	FR-03 – FR-09
Process / Methods	No tracking or feedback to reporter	FR-08, FR-10, FR-17
Management / Policy	Unclear responsibility; weak enforcement	FR-18, FR-17 (logged accountability)

Management / Policy	No anonymity / confidentiality policy	FR-05, NFR-03, NFR-07
Technology	No dedicated platform; records get lost	Whole system; NFR-12
Technology	No data for trend analysis	FR-14, FR-15, FR-22
Communication	Victims don't know who to contact; bodies don't share information	FR-01, FR-18
Environment	Scattered housing; insecurity	FR-06 (location, severity), FR-14 (emergencies)

10. Appendices

Appendix A – Questionnaire

A Google Form questionnaire was designed for both students and indigenes, branching after the first question so that each group answers questions relevant to them (level/faculty/residence for students; community/occupation/years of residence for indigenes) before common sections on issues experienced, current reporting channels and desired IRO features. See `docs/questionnaire/`.

Appendix B – Future Enhancements

- Server database and REST API; multiple officer accounts with roles.
- SMS / e-mail / WhatsApp status notifications; USSD reporting for feature phones.
- Photo and video evidence upload.
- Yoruba language option.
- Escalation authorities log in to respond directly.
- Monthly public report of resolved issues by category and community.